

Sai Kumar Vemula

DevSecOps Engineer | Cloud Security | Cybersecurity Analyst
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PROFESSIONAL SUMMARY

Security Engineer with 4+ years of experience spanning DevSecOps, AWS cloud security, and SOC-based cybersecurity operations. Proven track record building secure CI/CD pipelines, automating cloud security posture management, and reducing vulnerability exposure across multi-account AWS environments. Skilled in integrating SAST/DAST/SCA into pipelines, hardening Kubernetes and container workloads, and driving incident response, threat detection, and compliance programs (NIST, CIS, ISO 27001, PCI-DSS). Adept at scripting automation with Python and Bash, and collaborating with engineering, DevOps, and compliance teams to embed security into every stage of the SDLC. Open to DevSecOps, AWS/Cloud Security, Cybersecurity Analyst, Vulnerability Management, and SOC Analyst roles.

CORE SKILLS

Cloud Platforms: AWS (EC2, S3, IAM, VPC, Lambda, EKS, KMS, GuardDuty, Security Hub, Config, CloudTrail, WAF, Shield, Organizations/SCPs), Azure (basic), GCP (basic)

DevSecOps & CI/CD: Jenkins, GitHub Actions, GitLab CI, Azure DevOps, Terraform, CloudFormation, Ansible, Docker, Kubernetes, Helm, ArgoCD, Vault, OPA/Gatekeeper

Security Testing (SAST/DAST/SCA): SonarQube, Checkmarx, Veracode, Snyk, Trivy, tfsec, Checkov, OWASP ZAP, Burp Suite, Nessus, Qualys, Nikto

SIEM / SOC / EDR: Splunk, IBM QRadar, Microsoft Sentinel, CrowdStrike Falcon, Carbon Black, Wireshark, ELK Stack

Identity & Network Security: IAM/Least Privilege, MFA, SSO/SAML, Active Directory, Palo Alto, Fortinet, Cisco ASA, VPN, Zero Trust, Network Segmentation

Compliance & Frameworks: NIST 800-53, CIS Benchmarks, ISO 27001, PCI-DSS, SOC 2, HIPAA, MITRE ATT&CK, OWASP Top 10

Scripting & Automation: Python, Bash, PowerShell, YAML, JSON, REST APIs, Git

Monitoring & Vulnerability Mgmt: Prometheus, Grafana, Datadog, AWS Inspector, Rapid7 InsightVM, Patch Management, Threat Intelligence

Tools & Documentation: Microsoft Excel, Word, PowerPoint, SharePoint, Office 365, Adobe Acrobat

Soft Skills: Project Management, Critical Thinking, Analytical Skills, Communication, Attention to Detail, Initiative-Driven, Curiosity, Resilience

PROFESSIONAL EXPERIENCE

DevSecOps Engineer | Datadog — United States

June 2025 – Present

- Design and maintain secure CI/CD pipelines with Jenkins and GitHub Actions, embedding automated security gates across build, test, and deploy for 20+ microservices.
- Integrate SAST (SonarQube, Checkmarx) and SCA (Snyk) into pipelines, cutting critical/high vulnerabilities reaching production by 60%+.
- Enforce container image scanning with Trivy and a zero-critical policy before images promote to the production ECR registry.
- Automate IaC security checks with tfsec and Checkov to catch Terraform misconfigurations pre-deployment.
- Build OPA/Gatekeeper policies and harden Amazon EKS via network policies, Pod Security Standards, and RBAC least-privilege.
- Deploy HashiCorp Vault for centralized secrets management, eliminating hardcoded credentials from code and pipelines.
- Configure AWS GuardDuty, Security Hub, and Config for continuous CSPM aligned to the CIS AWS Foundations Benchmark (95%+ compliance); automate remediation with Lambda and Python.
- Run DAST (OWASP ZAP) in staging and lead STRIDE threat-modeling sessions with dev teams as a security champion.
- Manage least-privilege IAM (70% fewer wildcard policies), WAF rules, and AWS Shield DDoS protection for public-facing apps.
- Centralize logging with CloudTrail, CloudWatch, and the ELK stack; automate SOC 2 and ISO 27001 evidence collection, cutting audit prep by 40%.

Environment: AWS (EKS, Lambda, IAM, GuardDuty, Security Hub, Config, CloudTrail, KMS, WAF, Shield), Terraform, Jenkins, GitHub Actions, Docker, Kubernetes, HashiCorp Vault, SonarQube, Checkmarx, Snyk, Trivy, tfsec, Checkov, OPA/Gatekeeper, OWASP ZAP, Python, Bash, ELK Stack

- Administered a multi-account AWS Organization, enforcing org-wide guardrails with Service Control Policies (SCPs).
- Designed least-privilege IAM for 150+ users/roles, removing unused permissions flagged by IAM Access Analyzer.
- Deployed and tuned GuardDuty and Security Hub across all accounts into a centralized triage dashboard.
- Implemented AWS Config rules to assess compliance against CIS AWS Foundations Benchmark and internal baselines.
- Enforced encryption at rest/in transit with AWS KMS across S3, EBS, RDS, and DynamoDB.
- Built VPC security (security groups, NACLs, private subnets, endpoints) plus AWS WAF and Shield Advanced for public apps.
- Automated remediation with Lambda and Python (auto-quarantine of exposed S3 buckets and unencrypted volumes).
- Centralized CloudTrail and CloudWatch logging into Splunk for correlation and alerting.
- Standardized infrastructure baselines with Terraform and CloudFormation; managed Secrets Manager and Parameter Store for credential rotation.
- Ran AWS Well-Architected security reviews and prepared evidence for PCI-DSS and ISO 27001 audits.

***Environment:** AWS (IAM, GuardDuty, Security Hub, Config, CloudTrail, KMS, WAF, Shield, VPC, Organizations/SCPs, Secrets Manager), Terraform, CloudFormation, Splunk, Python, AWS Well-Architected Framework, CIS Benchmarks*

Cybersecurity Engineer | NSEIT— Mumbai, India

January 2021 – February 2023

- Monitored and triaged 50+ daily alerts in a 24x7 SOC using Splunk and IBM QRadar SIEM.
- Performed Tier 1/2 incident response for phishing, malware, and unauthorized access per documented playbooks.
- Ran vulnerability assessments with Nessus and Qualys, prioritizing remediation by CVSS severity.
- Managed Windows/Linux patch management within SLA windows and tracked cases in ServiceNow.
- Administered and tuned CrowdStrike Falcon EDR, reducing false positives by 30%.
- Configured firewall rules on Palo Alto and Fortinet for network segmentation and least-access policies.
- Correlated logs across firewalls, IDS/IPS, endpoints, and Active Directory to identify IOCs; used Wireshark for traffic analysis.
- Applied MITRE ATT&CK mapping to improve detection coverage and threat intelligence sharing.
- Implemented DLP policies and ran phishing simulations, improving employee report rates by 45%.
- Automated log parsing, alert enrichment, and reporting with Python and Bash; supported ISO 27001 and PCI-DSS audits.

***Environment:** Splunk, IBM QRadar, CrowdStrike Falcon, Nessus, Qualys, Palo Alto, Fortinet, Active Directory, Windows Server, Linux, Wireshark, ServiceNow, Python, Bash, MITRE ATT&CK*

EDUCATION

Yeshiva University, Katz School of Science and Health | New York
Master of Science in Cybersecurity | 3.9 CGPA

Aug 2024-May 2026

CERTIFICATIONS

- CompTIA Security + CompTIA
- CompTIA CySA+ (in progress)
- Google Cybersecurity Professional Certificate (Coursera)

Projects

Smart SIEM Risk Engine | Python, FastAPI, WebSockets, SQLite, Scikit-learn, MITRE ATT&CK

- Built an automated SIEM solution for security monitoring, log analysis, and threat detection, reducing SOC alert fatigue.
- Developed a machine learning-based risk scoring engine (Random Forest) for threat classification and alert prioritization.
- Implemented incident response automation and detection engineering, with real-time dashboards, GeoIP visualization, and alert notifications. GitHub <https://github.com/svemula17/smart-siem-risk-engine>

Agentic Penetration-Testing Assistant — Personal Project · Python, LangGraph, Claude/Gemini APIs, Docker

- Built an autonomous LLM agent that runs a full recon→exploitation pentest chain against a lab target, orchestrating 16 security tools via LangGraph with hard safety caps.
- Designed AI-agent guardrails — code-enforced target allowlist + sandboxed tool use + step limits — mitigating OWASP LLM Top 10 "Excessive Agency"; multi-LLM backend (Claude + Gemini) with failover and prompt caching.
- Demonstrated autonomous exploitation (agent-confirmed SQL injection + DB enumeration on DVWA) with auto-generated severity-ranked reports; 168 tests, ~7.4K LOC.